

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-0927

Helicon 138 kV
May 02, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0927. The Interconnection Customer proposes a 392.6 MW wind facility interconnecting at the Helicon 138 kV substation in the Pennsylvania area. The request was received on January 22, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 361 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 16 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$134,611,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: WATSONVILLE SUBSTATION – HUMBOLDT BAY 138.0 kV
Contingency: COMO – HELICON 138.0 kV
Loading: 241.9% (Rating: 158 MVA)
DFAX: 100.0%
Pre-project loading: 5.9%
2. Facility: HMBOLDT BAY – HELICON 138.0 kV
Contingency: COMO – HELICON 138.0 kV
Loading: 211.8% (Rating: 184 MVA)
DFAX: 100.0%
Pre-project loading: 2.0%
3. Facility: HELICON – COMO 138.0 kV
Contingency: HUMBOLDT BAY – HELICON 138.0 kV Line
Loading: 201.8% (Rating: 193 MVA)
DFAX: 100.0%
Pre-project loading: 1.9%
4. Facility: JARALES TAP SUBSTATION – COMO 138.0 kV L
Contingency: HUMBOLDT BAY – HELICON SUBSTATION 138.0 kV
Loading: 185.1% (Rating: 206 MVA)
DFAX: 100.0%
Pre-project loading: 5.4%
5. Facility: TEXACO TRADING – JARALES TAP 138.0 kV
Contingency: HUMBOLDT BAY – HELICON 138.0 kV
Loading: 174.4% (Rating: 215 MVA)
DFAX: 100.0%
Pre-project loading: 8.1%

6. Facility: BROOKINGS – WATSONVILLE 138.0 kV
Contingency: HELICON – COMO 138.0 kV
Loading: 167.8% (Rating: 223 MVA)
DFAX: 100.0%
Pre-project loading: 8.6%
7. Facility: TEXACO TRADING – PASTILLO 5801 138.0 kV
Contingency: HUMBOLDT BAY – HELICON 138.0 kV line
Loading: 166.1% (Rating: 221 MVA)
DFAX: 100.0%
Pre-project loading: 11.6%
8. Facility: PANDA PATRIOT GENERATION PLANT – FINLEY BUTTE 138.0 kV
Contingency: MONTOUR STA – TLENENERGY MARTINS CRK LLC WILLIAMSPORT 345.0 kV
Loading: 163.6% (Rating: 185 MVA)
DFAX: 11.4%
Pre-project loading: 139.5%
9. Facility: MONTOUR STATION 500/345 kV Transformer Bank
Contingency: CORONET – FT HUMBUG 500.0 kV
Loading: 138.2% (Rating: 883 MVA)
DFAX: 14.2%
Pre-project loading: 131.9%
10. Facility: ROCK SPRING – FINLEY BUTTE 138.0 kV
Contingency: TALENENERGY MARTINS CRK LLC WILLIAMSPORT – MONTOUR STATION SUBSTATION 345.0 kV
Loading: 137.9% (Rating: 206 MVA)
DFAX: 11.4%
Pre-project loading: 116.2%
11. Facility: PASTILLO 5801 138-1 kV Transformer
Contingency: MONTOUR STA – TALENENERGY MARTINS CRK LLC WILLIAMSPORT 345.0 kV
Loading: 125.7% (Rating: 319 MVA)
DFAX: 40.7%
Pre-project loading: 75.6%
12. Facility: BROOKINGS – COMPTON 138.0 kV
Contingency: PASTILLO 5801 345-1 kV Auto
Loading: 117.1% (Rating: 202 MVA)
DFAX: 50.4%
Pre-project loading: 19.4%
13. Facility: PASTILLO 5801 SUBSTATION 345 kV/1 kV Autotransformer
Contingency: MONTOUR STA – TALENENERGY MARTINS CRK LLC WILLIAMSPORT 345.0 kV
Loading: 108.7% (Rating: 369 MVA)
DFAX: 40.7%
Pre-project loading: 65.4%
14. Facility: JH WARDEN 500/345kV Autotransformer
Contingency: ACADIA – MONTOUR STATION 500.0 kV
Loading: 107.1% (Rating: 678 MVA)
DFAX: 5.3%
Pre-project loading: 104.1%
15. Facility: TALENENERGY MARTINS CRK LLC HARWOOD 345/1kV Autotransformer
Contingency: FELIDA 500-345 kV Auto
Loading: 104.4% (Rating: 1,482 MVA)
DFAX: 5.2%
Pre-project loading: 103.1%

16. Facility: TALENENERGY MARTINS CRK LLC HARWOOD 500-1 kV Transformer Bank Bank 1
 Contingency: FELIDA 500/345kV Transformer Bank
 Loading: 103.4% (Rating: 1,497 MVA)
 DFAX: 5.2%
 Pre-project loading: 102.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HUMBOLDT BAY – WATSONVILLE 138.0 kV	Rebuild line with higher-capacity conductor	\$11,488,000	100% (\$11,488,000)
HUMBOLDT BAY SUBSTATION – HELICON 138.0 kV	Rebuild line with higher-capacity conductor	\$11,376,000	100% (\$11,376,000)
HELICON – COMO 138.0 kV	Rebuild line with higher-capacity conductor	\$12,017,000	100% (\$12,017,000)
COMO SUBSTATION – JARALES TAP 138.0 kV	Rebuild line with higher-capacity conductor	\$12,835,000	100% (\$12,835,000)
TEXACO TRADING – JARALES TAP 138.0 kV	Rebuild line with higher-capacity conductor	\$12,858,000	100% (\$12,858,000)
BROOKINGS – WATSONVILLE 138.0 kV	Rebuild line with higher-capacity conductor	\$14,285,000	100% (\$14,285,000)
TEXACO TRADING – PASTILLO 5801 138.0 kV L	Rebuild line with higher-capacity conductor	\$2,575,000	100% (\$2,575,000)
PANDA PATRIOT GENERATION PLANT – FINLEY BUTTE 138.0 kV	Rebuild line with higher-capacity conductor	\$10,448,000	100% (\$10,448,000)
MONTOUR STATION 500/345 kV Transformer Bank	Replace transformer with higher-rated unit	\$29,411,000	18% (\$5,305,000)
ROCK SPRING – FINLEY BUTTE 138.0 kV	Rebuild line with higher-capacity conductor	\$3,083,000	100% (\$3,083,000)
PASTILLO 5801 138-1 kV Transformer	Replace transformer with higher-rated unit	\$6,649,000	100% (\$6,649,000)
BROOKINGS SUBSTATION – COMPTON 138.0 kV	Rebuild line with higher-capacity conductor	\$9,947,000	100% (\$9,947,000)
PASTILLO 5801 SUBSTATION 345 kV/1 kV Autotransformer	Replace transformer with higher-rated unit	\$12,374,000	100% (\$12,374,000)
JH WARDEN 500/345 kV TX	Replace transformer with higher-rated unit	\$28,795,000	18% (\$5,274,000)
TALENENERGY MARTINS CRK LLC HARWOOD 345-1 kV Transformer	Replace transformer with higher-rated unit	\$22,005,000	8% (\$1,806,000)
TALENENERGY MARTINS CRK LLC HARWOOD 500-1 kV Transformer Bank Bank 1	Replace transformer with higher-rated unit	\$27,908,000	8% (\$2,291,000)
	Total allocated to project		\$134,611,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.